Summarize

Part I. Foundational Concepts

Before we get started describing distributed systems, there are motivations and concepts that form the foundation of both why and how we build distributed systems. This section covers these foundational concepts to provide a basis for the rest of the book.

Distributed systems don't exist in a vacuum. The development of such systems is based on the evolving role of both computing and online systems in business and entertainment. In particular, in the development of always-on, mission-critical systems that we rely on every day. Additionally, the development of modern distributed systems is based on the history of how such systems have been designed and built in the past. This history of both how the systems are built, and often more importantly how they have failed, has led us to the current containerized and microservice architectures that you find in this book.

Before the design of distributed systems can be described, it is necessary to have a grounding in core concepts for how server systems operate, as well as fundamental computer science concepts like locking and APIs. It is also necessary to have a grounding in basic operations for distributed systems, such as monitoring and logging. Finally, because distributed systems involve numerous interactions across many different systems and many different requests, it is necessary to have a basic understanding of statistics and how we can measure the common behavior across the system through observing multiple different requests on different machines.

After reading these introductory chapters, you should have the foundational grounding in the context, history, and concepts necessary to understand how the design of these systems is described. This grounding also helps explain why some of the seemingly complex aspects of the design become necessary for reliability or scale.

Previous chapter

< Preface

Next chapter

1. Introduction >